

Equivalently, ρ_t is the probability density of the random vector

$$Z_t := e^{-t}X_0 + \sigma_t Z, \quad Z \sim \mathcal{N}(0, I_d),$$

where Z is independent from X_0 . Since $\mathbb{E}\|Z_t\|^2 = e^{-2t}M_2(P) + \sigma_t^2 d < \infty$, we have $\rho_t \in \mathcal{P}_2$ and this proves (i).

To prove (ii): Because the law of (Z_t, X_0) is a coupling of ρ_t and P ,

$$\begin{aligned} \mathcal{W}_2(\rho_t, P)^2 &\leq \mathbb{E}\|Z_t - X_0\|^2 \\ &= \mathbb{E}\|(e^{-t} - 1)X_0 + \sigma_t Z\|^2 \\ &= (1 - e^{-t})^2 M_2(P) + (1 - e^{-2t})d \\ &\leq t^2 M_2(P) + 2td, \end{aligned}$$

where in the last inequality we used that $1 - e^{-x} \leq x$, $\forall x \geq 0$. Since $M_2(P) < \infty$, we have bounded $\mathcal{W}_2(\rho_t, P)^2$ to be $O(t)$. $\square$

Proof of Corollary 5.3. For the ε in Assumption 1, the existence of δ to make $\mathcal{W}_2(\rho_\delta, P) < \varepsilon$ is by Lemma C.1, and we also have $\rho_\delta \in \mathcal{P}_2^*$. The rest of the proof is the same as in Corollary 5.2 by starting from $p_0 = \rho_\delta$. $\square$

C.3 Proofs and lemmas in Section 5.2

Lemma C.2 (Lipschitz bound of ODE solution map). *Suppose for $\gamma > 0$, $\hat{v}(x, t)$ is C^1 in (x, t) and Lipschitz in x uniformly on $\mathbb{R}^d \times [0, \gamma]$ with Lipschitz constant $K \geq 0$. Let $x(t)$ be the solution to the ODE*

$$\dot{x}(t) = \hat{v}(x(t), t), \quad t \in [0, \gamma], \quad (\text{A.22})$$

and define the solution map from 0 to γ as $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$, that is,

$$T(x_0) = x_0 + \int_0^\gamma \hat{v}(x(t), t) dt, \quad x(0) = x_0. \quad (\text{A.23})$$

Then T is invertible on $\mathbb{R}^d$, and both T and T^{-1} are Lipschitz on $\mathbb{R}^d$ with Lipschitz constant $e^{\gamma K}$.

Proof of Lemma C.2. Let $x_1(t)$ and $x_2(t)$ be the solution to the ODE (A.22) from $x_1(0) = y$, and $x_2(0) = z$ respectively. By definition,

$$T(y) = x_1(\gamma), \quad T(z) = x_2(\gamma).$$

Under the condition of $\hat{v}$, the ODE is well-posed [64]. This implies the invertibility of T , and T^{-1} is the solution map of the reverse time ODE from $t = \gamma$ to $t = 0$.

We now prove the Lipschitz constant of T on $\mathbb{R}^d$, and that of T^{-1} can be proved similarly by considering the reverse time ODE. We want to show that

$$\|T(y) - T(z)\| \leq e^{\gamma K} \|y - z\|, \quad \forall y, z \in \mathbb{R}^d,$$

and this is equivalent to that for any $x_1(0), x_2(0) \in \mathbb{R}^d$,

$$\|x_1(\gamma) - x_2(\gamma)\| \leq e^{\gamma K} \|x_1(0) - x_2(0)\|. \quad (\text{A.24})$$

For fixed $x_1(0), x_2(0)$, define

$$E(t) := \frac{1}{2} \|x_1(t) - x_2(t)\|^2,$$

then $E(0) = \|x_1(0) - x_2(0)\|^2/2$, and

$$\begin{aligned}\dot{E}(t) &= (x_1(t) - x_2(t))^T (\dot{x}_1(t) - \dot{x}_2(t)) \\ &= (x_1(t) - x_2(t))^T (\hat{v}(x_1(t), t) - \hat{v}(x_2(t), t)).\end{aligned}$$

Thus, by that $\|\hat{v}(x_1(t), t) - \hat{v}(x_2(t), t)\| \leq K\|x_1(t) - x_2(t)\|$, we have

$$\dot{E}(t) \leq K\|x_1(t) - x_2(t)\|^2 = 2KE(t).$$

By Grönwall's inequality, $E(t) \leq E(0)e^{2Kt}$, and this gives

$$\|x_1(t) - x_2(t)\|^2 \leq e^{2Kt}\|x_1(0) - x_2(0)\|^2, \quad t \in [0, \gamma].$$

Setting $t = \gamma$ proves (A.24). □

Lemma C.3. Suppose $p \in \mathcal{P}_2^r$, and $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is Lipschitz on $\mathbb{R}^d$, then $T \in L^2(p)$.

Proof. We are to show that

$$\mathbb{E}_{x \sim p} \|T(x)\|^2 < \infty.$$

Suppose T is L -Lipschitz on $\mathbb{R}^d$, then $\forall x \in \mathbb{R}^d$,

$$\|T(x)\| \leq \|T(0)\| + \|T(x) - T(0)\| \leq \|T(0)\| + L\|x\|.$$

Thus,

$$\begin{aligned}\mathbb{E}_{x \sim p} \|T(x)\|^2 &\leq 2(\|T(0)\|^2 + L^2 \mathbb{E}_{x \sim p} \|x\|^2) \\ &= 2(\|T(0)\|^2 + L^2 M_2(p)) < \infty,\end{aligned}$$

because $M_2(p) < \infty$. □

Proof of Proposition 5.4. By construction, for $n = 1, \dots, N$,

$$q_{n-1} = (T_n^{-1})_{\#} q_n, \quad \tilde{q}_{n-1} = (S_n)_{\#} \tilde{q}_n.$$

We also have that $q_N = \tilde{q}_N = q \in \mathcal{P}_2^r$ by assumption.

For the sequence of q_n , we know that T_n^{-1} is non-degenerate (Assumption 1). Meanwhile, for $q_n \in \mathcal{P}_2^r$, T_n^{-1} being globally Lipschitz on $\mathbb{R}^d$ also implies that it is in $L^2(q_n)$ (Lemma C.3). Then by Lemma 3.4, $q_{n-1} = (T_n^{-1})_{\#} q_n$ is also in $\mathcal{P}_2^r$. For the sequence of $\tilde{q}_n$, by Assumption 2, S_n is non-degenerate and in $L^2(\tilde{q}_n)$, thus from $\tilde{q}_n \in \mathcal{P}_2^r$, $\tilde{q}_{n-1} = (S_n)_{\#} \tilde{q}_n$ is also in $\mathcal{P}_2^r$ by Lemma 3.4 again. Thus by induction, we have that q_n and $\tilde{q}_n$ are all in $\mathcal{P}_2^r$.

For each n , we have

$$\begin{aligned}\mathcal{W}_2(\tilde{q}_{n-1}, q_{n-1}) &= \mathcal{W}_2((S_n)_{\#} \tilde{q}_n, (T_n^{-1})_{\#} q_n) \\ &\leq \underbrace{\mathcal{W}_2((S_n)_{\#} \tilde{q}_n, (T_n^{-1})_{\#} \tilde{q}_n)}_{\textcircled{1}} \\ &\quad + \underbrace{\mathcal{W}_2((T_n^{-1})_{\#} \tilde{q}_n, (T_n^{-1})_{\#} q_n)}_{\textcircled{2}}.\end{aligned}$$

To bound ①, we use Assumption 2 and the Lipschitzness of T_n^{-1} . Define $L := e^{\gamma K}$. Using $(S_n, T_n^{-1})_{\#} \tilde{q}_n$ as the coupling, we have

$$\begin{aligned} & \mathcal{W}_2^2((S_n)_{\#} \tilde{q}_n, (T_n^{-1})_{\#} \tilde{q}_n) \\ & \leq \int_{\mathbb{R}^d} \|S_n(x) - T_n^{-1}(x)\|^2 \tilde{q}_n(x) dx \\ & \leq \int_{\mathbb{R}^d} L^2 \|T_n \circ S_n(x) - x\|^2 \tilde{q}_n(x) dx \\ & = L^2 \|T_n \circ S_n - \text{Id}\|_{\tilde{q}_n}^2, \end{aligned}$$

where for the second inequality, we use the fact that T_n^{-1} is L -Lipschitz. Thus,

$$\textcircled{1} \leq L \varepsilon_{\text{inv}}. \quad (\text{A.25})$$

To bound ②, we use that T_n^{-1} is L -Lipschitz on $\mathbb{R}^d$ again. Specifically, let Y_n be the unique OT map from q_n to $\tilde{q}_n$ which is well-defined by the Brenier Theorem, then $(T_n^{-1} \circ Y_n, T_n^{-1})_{\#} q_n$ is a coupling of $(T_n^{-1})_{\#} \tilde{q}_n$ and $(T_n^{-1})_{\#} q_n$. We have that

$$\begin{aligned} & \mathcal{W}_2^2((T_n^{-1})_{\#} \tilde{q}_n, (T_n^{-1})_{\#} q_n) \\ & \leq \int_{\mathbb{R}^d} \|T_n^{-1} \circ Y_n(x) - T_n^{-1}(x)\|^2 q_n(x) dx \\ & \leq \int_{\mathbb{R}^d} L^2 \|Y_n(x) - x\|^2 q_n(x) dx \\ & = L^2 \mathcal{W}_2^2(\tilde{q}_n, q_n). \end{aligned}$$

Thus

$$\textcircled{2} \leq L \mathcal{W}_2(\tilde{q}_n, q_n). \quad (\text{A.26})$$

Putting together, we have

$$\mathcal{W}_2(\tilde{q}_{n-1}, q_{n-1}) \leq e^{\gamma K} (\varepsilon_{\text{inv}} + \mathcal{W}_2(\tilde{q}_n, q_n)).$$

Note that $\mathcal{W}_2(\tilde{q}_N, q_N) = 0$ by that $\tilde{q}_N = q_N = q$. Applying recursively from $n = N$ to $n = 1$ gives that

$$\mathcal{W}_2(\tilde{q}_0, q_0) \leq \varepsilon_{\text{inv}} \frac{e^{\gamma K} (e^{\gamma K N} - 1)}{e^{\gamma K} - 1}, \quad (\text{A.27})$$

which proves (41) by that $e^x - 1 \geq x$ for any $x \in \mathbb{R}$. $\square$

Proof of Corollary 5.5. Under the condition of the corollary, Corollary 5.2 applies to bound $\text{KL}(p||q_0)$ and $TV(p, q_0)$ as in (40), and Proposition 5.4 applies to bound $\mathcal{W}_2(\tilde{q}_0, q_0)$ as in (41). It suffices to show that the r.h.s. of (41) is less than or equal to that of (42).

By the choice of N in (39),

$$N \leq \frac{8}{\gamma \lambda} (\log \mathcal{W}_2(p_0, q) + \log(\lambda/\varepsilon)) + 1,$$

and thus

$$e^{\gamma K(N+1)} \leq e^{2\gamma K} \left(\mathcal{W}_2(p_0, q) \frac{\lambda}{\varepsilon} \right)^{8K/\lambda},$$

which proves the needed inequality. $\square$